Here is a structured 12-week learning roadmap designed for a beginner committing **5–7 hours per week**. This path moves systematically from core spreadsheet concepts to SQL, basic statistics, Python with Pandas, and data visualization, ending with a complete portfolio project.

Access & Pricing Guide

Resource / Platform	Access Level & Pricing Model
Kaggle Learn	**100% Free**: Tutorials, coding environments, and micro-certificates.
SQLBolt	**100% Free**: Interactive, browser-based SQL exercises.
freeCodeCamp / YouTube	**100% Free**: Open-access tutorials and full video courses.
Coursera	**Free Audit / Paid Cert**: Free to view course materials via the "Audit" option. Graded assignments, verified certificates, or full access require a monthly subscription or financial aid.
Excel / Google Sheets	**Free Option**: Google Sheets is free with a Google Account. Microsoft Excel requires a paid Microsoft 365 license or free web access.

12-Week Roadmap Overview



[Weeks 11–12] End-to-End Capstone Portfolio Project

Weekly Learning Roadmap Table

Week	Topics to Learn	Learning Objectives	Recommended Courses / Tutorials	Tools & Software	Practical Exercise / Mini-Project	Study Time	Measurable Completion Criterion
1	**Spreadsheet Basics & Data Cleaning**	Learn cell referencing, standard text/date formatting, sorting, filtering, and data cleaning techniques (TRIM, UPPER/LOWER, REMOVE DUPLICATES).	• Google Sheets Training Center *(Free)* • Coursera: Excel Basics for Data Analysis *(Free Audit / Paid Cert)*	Google Sheets or Microsoft Excel	Clean a messy sales dataset by removing duplicate rows, trimming standard spaces, fixing dates, and standardizing category names.	**5–6 hrs**	Successfully clean all errors in the practice dataset without losing valid rows.
2	**Spreadsheet Formul**	Master lookup functio	• freeCodeCam	Google Sheets or	Build a dynamic sales	**6–7 hrs**	Produce a single

	as & Pivot Tables	ns (VLOOKUP, XLOOKUP, INDEX/MATCH), logical functions (IF, SUMIFS, COUNTIFS), and summarize data using Pivot Tables and Pivot Charts.	p Excel/Sheets Tutorial *(Free)* • GCFGlobal Excel Tutorials *(Free)*	Microsoft Excel	performance summary using Pivot Tables and aggregate formulas (SUMIFS) to show monthly revenue by region.		dashboard tab with aggregated summaries and 2 dynamic charts.
3	**SQL Foundations**	Understand relational database concepts, tables, primary/foreign keys, and basic SELECT, WHERE, ORDER BY, and LIMIT	• SQLBolt: Lessons 1–5 *(Free)* • Mode Analytics SQL Tutorial - Basic *(Free)*	SQLBolt (Browser-based) or PostgreSQL	Query a movie or e-commerce dataset to filter and rank transactions based on price, region, and date thresholds.	**5–6 hrs**	Solve all interactive exercises in SQLBolt Lessons 1 through 5 with zero query errors.

		queries.					
4	**SQL Aggregations & Joins**	Perform group aggregations using GROUP BY, HAVING, COUNT, SUM, AVG, and combine multiple tables using INNER, LEFT, RIGHT, and FULL JOIN.	• SQLBolt: Lessons 6–12 *(Free)* • Kaggle Learn: Intro to SQL *(Free)*	SQLite Studio, DBeaver, or Browser SQL	Write multi-table join queries to calculate total order revenue per customer and filter out groups with low sales volume.	**6–7 hrs**	Successfully execute 5 complex queries combining INNER JOIN and GROUP BY on a sample database.
5	**Intermediate SQL & Data Extraction**	Master subqueries, CTEs (WITH clauses), conditional statements (CASE WHEN), and	• SQLBolt: Lessons 13–18 *(Free)* • Mode Analytics SQL Tutorial - Intermediate	PostgreSQL / pgAdmin or SQLite	Extract and classify customer orders into categories (e.g., "High", "Medium", "Low" value)	**6–7 hrs**	Export filtered SQL query output directly into a CSV file ready for analysis.

		basic string/date formatting in SQL queries.	*(Free)*		using CASE WHEN inside a CTE.		
6	**Basic Statistics for Data Analysis**	Understand descriptive statistics: mean, median, mode, standard deviation, variance, percentiles, interquartile range (IQR), and spotting outliers.	• Khan Academy: Descriptive Statistics *(Free)* • Coursera: Basic Statistics *(Free Audit)*	Google Sheets or Python (Jupyter Notebook)	Calculate summary statistics and detect extreme outliers in an e-commerce pricing dataset.	**5–6 hrs**	Write a 1-page summary explaining the skewness and outlier presence in the target dataset.
7	**Data Visualization & Dashboarding**	Learn data visual design rules, choosing charts	• Tableau Public Resources *(Free)*	Tableau Public or Power BI Desktop (Free)	Design an interactive 3-chart sales dashbo	**6–7 hrs**	Publish a working interactive dashboard to

		(bar, line, scatter, boxplot), and creating interactive dashboards with filters and drill-downs.	• Coursera: Data Visualization with Tableau *(Free Audit)*		ard featuring continuous filters for regions, years, and product lines.		Tableau Public or export an interactive Power BI report.
8	**Python Fundamentals for Data**	Learn Python data types (strings, lists, dictionaries), control flow (if/else), loops (for/while), and basic function declarations.	• Kaggle Learn: Python *(Free)* • freeCodeCamp: Scientific Computing with Python *(Free)*	Python 3, Jupyter Notebook, Google Colab	Write basic Python functions to iterate through a list of customer order dictionaries and return valid order totals.	**5–6 hrs**	Complete all Kaggle Python micro-course exercises or equivalent Google Colab exercises.
9	**Data Wrangling with Pandas**	Learn to load datasets	• Kaggle Learn: Pandas	Jupyter Notebook or Google	Load an uncleaned CSV dataset,	**6–7 hrs**	Run a script without syntax

		(.read_csv), inspect DataFrames (.head(), .info(), .describe()), handle missing data (.fillna(), .dropna()), and filter rows.	*(Free)* • Pandas Official Getting Started Guide *(Free)*	Colab	rename columns, fill missing numeric values with medians, and drop invalid rows.		or indexing errors that returns a clean Pandas DataFrame.
10	**Exploratory Data Analysis (EDA) in Python**	Perform grouping and aggregation (.groupby()), merging DataFrames (.merge()), and generate EDA plots using Seaborn/Matplotlib.	• Kaggle Learn: Data Visualization *(Free)* • Seaborn Official Tutorials *(Free)*	Python, Pandas, Matplotlib, Seaborn	Perform an exploratory analysis on an open dataset (e.g., HR attrition or retail store sales) and plot distributions/correlations.	**6–7 hrs**	Generate a clean Jupyter Notebook containing 4 distinct plots and concise analytical takeaways.
11	**Portfolio**	Define a	• Kaggle	SQL, Python,	Draft a project	**6–7 hrs**	Push clean

	Project: Planning & Data Preparation	business problem, source an open dataset (Kaggle/data.gov), perform data cleaning, and write SQL/Python queries to extract metrics.	Datasets *(Free)* • GitHub Guides: Documentation *(Free)*	Pandas, GitHub	README outlining problem statements, hypotheses, data sources, and complete initial data cleaning pipelines.		project code and raw/processed datasets into a public GitHub repository.
12	**Portfolio Project: Visualization & Presentation**	Build an interactive dashboard or complete visual story, document key insights, write a project conclusion, and publish a complete portfoli	• Data Science Portfolio Tips (freeCodeCamp) *(Free)*	GitHub, Tableau Public, Google Colab	Complete the final portfolio project report including problem definition, methodology, charts, findings, and business	**6–7 hrs**	Share a public GitHub link and dashboard link with a complete README document suitable for job applications.

		o reposit ory.			recom mendati ons.		

Skills Acquired After 12 Weeks

Upon successfully completing this 12-week roadmap, a learner will be able to demonstrate the following core competencies:

1. Data Cleaning & Manipulation

- Identify and clean dirty datasets (missing values, duplicates, bad date formats, inconsistent string entries) across both Spreadsheets and Python.
- Use advanced Excel/Sheets formulas (VLOOKUP, XLOOKUP, INDEX/MATCH, nested IF statements) and dynamic Pivot Tables.

2. Database Querying (SQL)

- Write queries using SELECT, WHERE, ORDER BY, GROUP BY, and HAVING filters.
- Combine data across multiple tables using INNER JOIN, LEFT JOIN, RIGHT JOIN, and subqueries/CTEs (WITH clauses).

3. Data Analysis & Statistics (Python & Pandas)

- Calculate and interpret key descriptive statistics (mean, median, standard deviation, interquartile ranges) to uncover data trends and anomalies.
- Process, filter, reshape, and aggregate large datasets using Pandas DataFrames.

4. Data Visualization & Storytelling

- Build clear visual representations (bar charts, line graphs, scatter plots, boxplots) using Seaborn/Matplotlib and Tableau Public.
- Design interactive dashboards tailored to business stakeholders.

5. Project Documentation & Version Control

- Set up public project repositories on **GitHub** with clean documentation (README.md), code commentary, and structured data pipelines to showcase to potential employers.